Leander Girrbach

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Education

2023 – Present **Ph.D. in Computer Science**
Technical University of Munich (TUM)
Research on large-scale empirical analysis of multimodal foundation models, efficient adaptation methods, and mechanistic interpretability. Advised by Prof. Zeynep Akata.

2021 – 2023 **MA in Computational Linguistics**
University of Tübingen
Thesis: *Autoregressive and Non-Autoregressive Character Level String Transduction with Hard Attention Neural Models.* Supervised by Dr. Çağrı Çöltekin. Grade: 1,0 (best possible)

2018 – 2021 **BA in Computational Linguistics**
Heidelberg University
Thesis: *Unsupervised contrastive learning of context sensitive embeddings from composition functions for English and German noun compounds.* Supervised by Prof. Annette Frank. Grade: 1,0

Research Experience (pre-PhD)

2023 **Student Research Assistant**
EML, University of Tübingen
Researched applications of Topological Data Analysis to neural network generalization. Supervised by Dr. Almut Sophia Koepke and Prof. Zeynep Akata. Published in TMLR 2023.

2022 – 2023 **Student Research Assistant**
Seminar für Sprachwissenschaft, University of Tübingen
Researched methods for domain-specific vocabulary extraction. Supervised by Prof. Detmar Meurers. Published in BEA@ACL 2025.

Grants and Fellowships

2025 – 2027 **Gauss Centre for Supercomputing (GCS)**
1.5 M GPU-hours for large-scale vision-language research (PI: Prof. Zeynep Akata; co-written proposal).

2017 – 2023 **Studienstiftung des Deutschen Volkes**
Financial and academic support. Awarded to top 0.5% of German university students.

Teaching Experience

2024 – 2025 Seminar *Advanced Topics in Vision-Language Models.* 2 iterations.
2023 (Summer) Lecture *Introduction to Machine Learning.* Teaching Assistant.
2021 (Summer) Lecture *Formal Foundations – Mathematical Foundations.* Teaching Assistant.
2020 (Winter) Lecture *Statistical Methods.* Teaching Assistant.
2020 (Summer) Lecture *Formal Foundations – Mathematical Foundations.* Teaching Assistant.
2019 (Winter) Lecture *Formal Foundations – Logical Foundations.* Teaching Assistant.

Research Infrastructure & Administration

2025 – Present	Project Administrator for GPU computing allocations on JUWELS Booster and JUPITER Booster (Jülich Supercomputing Centre).
2025 – Present	IT Representative of the EML lab. Liaison to Helmholtz Munich HPC management and cross-lab IT coordination.
2025 – Present	System administrator for lab-internal GPU servers.
2024 – Present	Project Administrator for LRZ AI Systems (Leibniz-Rechenzentrum, TU Munich).

Skills

ML Research	PyTorch, HuggingFace Transformers, vLLM, TRL/PEFT
Classical ML	numpy, scipy, pandas, scikit-learn
Infrastructure	HPC/Supercomputing, Python, Git, Linux, Docker
Languages	German (native), English (fluent), Italian (conversational), Japanese (conversational), Spanish/French (reading), Mandarin Chinese (elementary)

Thesis Supervision

2024–2025	Philipp Spohn. *Multi-Token Embedding Prediction in Autoregressive Language Models.* BSc in Computer Science. **(completed)** co-supervised with Jessica Bader
2024–2025	Utkarsh Gupta. *Probing the Linearity of LLMs with Affine Optimal Transport.* MSc in Computer Science. **(completed)** co-supervised with Quentin Bouniot
2025	Alexander von Recum. *Investigating the Robustness of Reasoning Models.* BSc in Computer Science. **(completed)**

Ongoing:

2025 –	Karim Lotfy. *Can Uncertainty Predict Correctness and Quality of LLM Generations?.* MSc in Computer Science.
2025 –	Mohamed Helmy. *Inspecting Medical Foundation Models with SAEs.* MSc in Computer Science. co-supervised with Quentin Bouniot and Furkan Daşdelen
2025 –	Nils Herrmann. *Investigating Reasoning Dynamics in Large Language Models through the Lens of Personas.* MSc in Computer Science. co-supervised with Kirill Bykov

Reviewing

2026	CVPR, ICLR, TMLR, ICML
2025	CVPR **(Outstanding Reviewer, top 6%)**, NeurIPS **(Top Reviewer, top 8%)**, ICML, ICLR, UAI, TMLR, ACL RR, WACV, ICCV Workshop eXCV, NeurIPS Workshop NeurReps
2024	ACL RR, UAI, GCPR

Publications

6 peer-reviewed publications (2× first-author at ICLR), 4 preprints, 4+ workshop papers.

† denotes supervised student project

Peer-Reviewed Publications

Leander Girrbach, Stephan Alaniz, Genevieve Smith, Trevor Darrell, and Zeynep Akata. *Person-Centric Annotations of LAION-400M: Auditing Bias and Its Transfer to Models.* ICLR, 2026.

Leander Girrbach, Stephan Alaniz, Yiran Huang, Trevor Darrell, and Zeynep Akata. *Revealing and Reducing Gender Biases in Vision and Language Assistants (VLAs).* ICLR, 2025.

Alexander von Recum[†] and **Leander Girrbach**, and Zeynep Akata. *Are Reasoning LLMs Robust to Interventions on their Chain-of-Thought?.* ICLR, 2026.

Massimo Bini, **Leander Girrbach**, and Zeynep Akata. *Decoupling Angles and Strength in Low-Rank Adaptation.* ICLR, 2025.

Jessica Bader, **Leander Girrbach**, Stephan Alaniz, and Zeynep Akata. *Sub: Benchmarking CBM Generalization via Synthetic Attribute Substitutions.* ICCV, 2025.

Leander Girrbach, Anders Christensen, Ole Winther, Zeynep Akata, and A. Sophia Koepke. *Addressing Caveats of Neural Persistence with Deep Graph Persistence.* TMLR, 2023.

Preprints & Under Review

Leander Girrbach and Zeynep Akata. *Sparse Autoencoders are Topic Models.* Under review, 2025.

Leander Girrbach, Stephan Alaniz, Genevieve Smith, and Zeynep Akata. *A Large Scale Analysis of Gender Biases in Text-to-Image Generative Models.* Under review, 2025.

Oğuz Kağan Hitit[†], **Leander Girrbach**, and Zeynep Akata. *A Systematic Study of Model Merging Techniques in Large Language Models.* Under review, 2025.

Leander Girrbach, Chi-Ping Su, Tankred Saanum, Richard Socher, Eric Schulz, and Zeynep Akata. *Reference-Free Rating of LLM Responses via Latent Information.* Preprint, 2025.

Workshop Publications

Philipp Spohn[†], **Leander Girrbach**, Jessica Bader, and Zeynep Akata. *Align-then-Unlearn: Embedding Alignment for LLM Unlearning.* ICML Workshop on Machine Unlearning for Generative AI, 2025.

Kordula De Kuthy, **Leander Girrbach**, and Detmar Meurers. *Automatic Concept Extraction for Learning Domain Modeling: A Weakly Supervised Approach Using Contextualized Word Embeddings.* ACL BEA Workshop, 2025.

Leander Girrbach. *Straight-Through Gradient Estimation for Hard Attention.* ACL SIGMORPHON, 2023.

Leander Girrbach et al.. *Multiple shared task submissions on morphological inflection, word segmentation, grapheme-to-phoneme conversion, and legal text structuring.* ACL SIGMORPHON, SemEval, IberLEF, 2022–2023.